

UNIT III – FIBER OPTICS

PART A – Apply / Analyze (2 Marks Each)

1. Derive the relation between Numerical Aperture (NA) and Acceptance Angle (θ_a) for an optical fiber.
 2. A fiber has a core refractive index of 1.5 and cladding index of 1.45. Calculate its NA and acceptance angle in air.
 3. Explain how total internal reflection enables light propagation through optical fibers.
 4. A light ray enters a silica fiber ($n_1 = 1.48$, $n_2 = 1.46$). Determine the maximum acceptance angle for total internal reflection.
 5. Discuss how attenuation and dispersion affect the performance of fiber optic communication.
 6. Explain why optical fibers are immune to electromagnetic interference (EMI).
 7. Apply Snell's law to derive the critical angle condition for fiber light confinement.
 8. A fiber has $n_1 = 1.46$ and $n_2 = 1.40$. Calculate its numerical aperture.
 9. How does the acceptance cone depend on the refractive indices of the core and cladding?
 10. A light ray enters the fiber at 25° to the axis. If $n_1 = 1.50$, find the minimum n_2 required for total internal reflection.
 11. Analyze the significance of cladding in an optical fiber.
 12. Describe one practical application of optical fibers in medicine and explain the underlying physics.
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PART B – Apply / Analyze

1. Calculate the numerical aperture, acceptance angle, and critical angle for a glass fiber with $n_1 = 1.52$ and $n_2 = 1.47$.
 2. Analyze signal loss mechanisms in optical fibers – absorption, scattering, and bending losses – and propose engineering methods to minimize them.
 3. A fiber system uses light of wavelength 850 nm, NA = 0.20, and core diameter = 50 μm . Calculate the number of guided modes supported.
 4. Discuss the advantages, challenges, and recent developments in optical fiber communication over metallic cables.
 5. Explain the construction and working principle of optical fibers. Analyze the role of total internal reflection and derive expressions for NA and acceptance angle.
 6. Explain with a schematic how optical fiber communication systems transmit information from transmitter to receiver.
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UNIT IV – SEMICONDUCTORS AND OPTOELECTRONIC DEVICES

PART A – Apply / Analyze (2 Marks Each)

1. Compare intrinsic and extrinsic semiconductors in terms of Fermi level and carrier concentration.
2. Analyze how temperature affects the conductivity of an intrinsic semiconductor.
3. Draw and interpret the I–V characteristics of a PN junction diode.
4. Distinguish between n-type and p-type semiconductors with energy band diagrams.
5. A Ge diode has barrier potential = 0.3 V; if forward current = 2 mA at 0.4 V, calculate current at 0.5 V ($\eta = 1$).
6. Explain how recombination of charge carriers in an LED produces light.
7. Analyze the working of a BJT in the active region and its application as an amplifier.
8. Explain the principle of photo-detection in photodiodes and solar cells.
9. Calculate the efficiency of a solar cell if input power = 100 mW/cm² and output = 15 mW/cm².
10. Analyze how doping concentration affects the Fermi level in n-type and p-type semiconductors.
11. Calculate the conductivity of a semiconductor with $n = 8 \times 10^{15} \text{ cm}^{-3}$, $\mu = 1300 \text{ cm}^2/Vs$, $q = 1.6 \times 10^{-19} \text{ C}$.
12. Distinguish between direct and indirect band gap semiconductors.
13. Draw and interpret the energy band diagram of a forward-biased diode.
14. A silicon diode has $I_S = 10^{-12} \text{ A}$. Calculate current when forward voltage = 0.7 V at 300 K.
15. Explain the working principle of a transistor in common emitter configuration.
16. Analyze how light intensity affects the current output in a photodiode.
17. Compare LEDs and solar cells in terms of energy conversion mechanism.
18. Determine the output power of a solar cell with current = 25 mA and voltage = 0.55 V.
19. Explain why silicon (Si) is preferred over germanium (Ge) in semiconductor devices.

PART B – Apply / Analyze (13 Marks Each)

1. Derive the current–voltage relationship of a PN junction diode and discuss the role of the ideality factor.
2. Analyze the energy band diagrams of intrinsic, n-type, and p-type semiconductors and explain carrier movement under bias.
3. A Si diode has $I_S = 10^{-12} \text{ A}$, $\eta = 1$. Find forward current at 0.6 V and 300 K ($kT/q = 26 \text{ mV}$).
4. Compare and analyze the operation principles of LED, photodiode, and solar cell with energy band diagrams.

5. Discuss the BJT operation modes (active, cutoff, saturation) and analyze its behavior as a switch in digital circuits.
 6. Design an LED-based optical transmitter circuit for a short-distance communication system and explain its working.
 7. A silicon diode has $I_S = 5 \times 10^{-13} \text{ A}$. Calculate current for 0.6 V and 0.7 V forward biases.
 8. Describe the operation of a BJT transistor and analyze its use as an amplifier.
 9. Evaluate how band gap engineering influences the performance of optoelectronic devices.
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UNIT V – NANOMATERIALS

PART A – Apply / Analyze (2 Marks Each)

1. Compare nanomaterials and bulk materials in terms of surface area and electrical properties.
 2. Explain how the quantum confinement effect influences the band gap in nanoparticles.
 3. Analyze the difference between nanoparticles, nanoclusters, and nanorods with examples.
 4. Why does the melting point of nanoparticles decrease with size?
 5. Distinguish between sol–gel and solvo-thermal synthesis.
 6. Explain how carbon nanotube (CNT) structure (single-wall vs multi-wall) affects conductivity.
 7. Analyze the advantages of electrospinning in nanowire fabrication.
 8. Compare electrochemical deposition and CVD for nanostructure growth.
 9. A gold nanoparticle ($r = 5\text{ nm}$) – estimate surface-to-volume ratio and compare with bulk.
 10. Discuss how nanomaterials are used in biosensing or drug delivery.
 11. Analyze how surface-to-volume ratio affects physical properties of nanomaterials.
 12. Calculate the percentage of surface atoms in a cubic nanoparticle of size 5 nm (atomic diameter $= 0.25\text{ nm}$).
 13. Explain why nanomaterials show enhanced catalytic activity.
 14. Compare top-down and bottom-up approaches.
 15. Distinguish between nanorods and nanowires in aspect ratio and behavior.
 16. Analyze how solvo-thermal synthesis controls particle morphology.
 17. Describe one sensor application of CNTs and explain its working.
 18. Compare the mechanical strength of CNTs with steel and explain.
 19. Explain how quantum confinement alters optical properties.
 20. A nanowire's resistance decreases from $200\text{ k}\Omega$ to $160\text{ k}\Omega$ upon gas exposure. Calculate % change in resistance and interpret.
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PART B – Apply / Analyze (13 Marks Each)

1. Explain the sol–gel process for synthesizing nanoparticles with schematic representation. Analyze the effect of pH and temperature on particle size.
2. Compare the structural, electrical, and mechanical properties of nanomaterials and bulk materials.
3. Analyze the CVD method for preparing CNTs and discuss their applications in nanoelectronics.
4. Explain the solvo-thermal synthesis method and how it enables control over morphology.

5. Describe the electrospinning process for producing nanowires and discuss how parameters affect fiber diameter.
6. Analyze how nanomaterials enhance sensor performance with examples (gas/bio sensors).
7. A nanowire sensor shows resistance change from $100\text{ k}\Omega$ to $75\text{ k}\Omega$ upon gas exposure. Calculate % sensitivity and interpret.
8. Discuss the optical and electrical applications of CNTs and nanowires in modern electronics.
9. Explain the electrochemical deposition method for nanowire fabrication and analyze the role of electrolyte concentration.
10. Analyze the applications of nanomaterials in electronics, sensors, and energy storage devices.
11. A nanowire-based sensor gives $80\text{ }\mu\text{A}$ at 1 ppm gas and $50\text{ }\mu\text{A}$ without gas. Compute % sensitivity and explain its significance.